

DECIMAL	HEXADECIMAL	DESCRIPTION
2322822	0x237186	ARM instructions, function prologue

Raw string

The '-R' option allows you to search for a custom string in the target file. The header string 'HDR0' is matched in the following example:

```
$ binwalk -R "HDR0" dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	Raw signature (HDR0)

Disabling signature match

You can disable the 'smart' signature match with the '-b' option as indicated below:

```
$ binwalk -b dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	TRX firmware header, little endian, image size: 3543040 bytes, CRC32: 0x85472C8C, flags: 0x0, version: 1, header size: 28 bytes, loader offset: 0x1C, linux kernel offset: 0x8E4, rootfs offset: 0x8E7CC
28	0x1C	gzip compressed data, maximum compression, from Unix, last modified: 1970-01-01 00:00:00 (null date)
2276	0x8E4	LZMA compressed data, properties: 0x5D, dictionary size: 8388608 bytes, uncompressed size: 1982464 bytes
583628	0x8E7CC	Squashfs filesystem, little endian, DD-WRT signature, version 3.0, size: 2954340 bytes, 587 inodes, blocksize: 65536 bytes, created: 2007-06-14 00:34:20
1674548	0x198D34	rzip compressed data - version 59.-98 (1020936805 bytes)

Entropy

The '-E' option performs entropy analysis on the target file and generates the entropy graph.

```
$ binwalk -E dd-wrt.v24_whr-g125.bin
```

DECIMAL	HEXADECIMAL	ENTROPY
0	0x0	Rising entropy edge (0.984773)
3536896	0x35F800	Falling entropy edge (0.626462)

You can also use the '--verbose' option for displaying a more detailed entropy calculation as follows:

```
$ binwalk -E --verbose dd-wrt.v24_whr-g125.bin
```

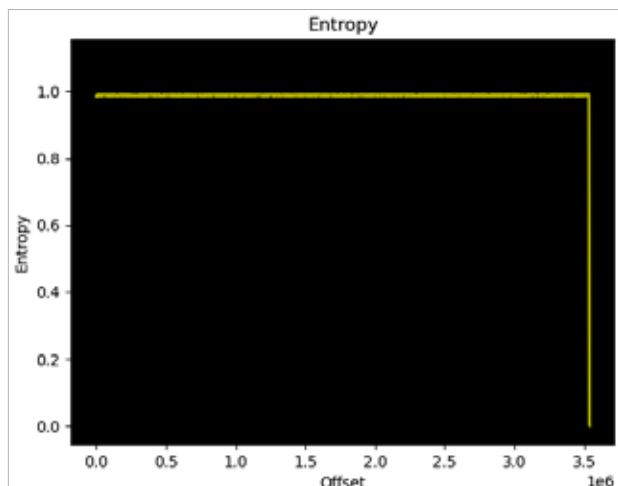


Figure 1: Entropy

Scan Time: 2024-08-18 21:06:24

Target File: /home/shakthi/operation-gladiator/osfy/security-series/5-binwalk/dd-wrt.v24_whr-g125.bin

MD5 Checksum: 8a60810685fa5be6221936034b81fd3a

MD5 Checksum: 8a60810685fa5be6221936034b81fd3a

DECIMAL	HEXADECIMAL	ENTROPY
0	0x0	0.984773
2048	0x800	0.988507
4096	0x1000	0.988300
6144	0x1800	0.989242
8192	0x2000	0.988839
10240	0x2800	0.989643
12288	0x3000	0.989161
14336	0x3800	0.987165
16384	0x4000	0.988030
18432	0x4800	0.989496
...		

Do read the binwalk GitHub documentation to learn more on its options, usage and API. **END** 🐧

References

- Kali binwalk documentation, <https://www.kali.org/tools/binwalk/>
- binwalk Quick Start Guide, <https://github.com/ReFirmLabs/binwalk/wiki/Quick-Start-Guide>
- binwalk usage, <https://github.com/ReFirmLabs/binwalk/wiki/Usage>
- binwalk API, <https://github.com/ReFirmLabs/binwalk/blob/master/API.md>
- DD WRT Buffalo WHR G125 firmware files, <https://dd-wrt.com/support/other-downloads/?path=obsolete%2Fbeta%2FBUFFALO+WHR-G125%2F2007+-+0614%2F>

By: Shakthi Kannan

The author is a free software enthusiast.